Huawei AI Solutions

Contents

1. **Huawei's Full-Stack, All-Scenario AI Solutions**

2. Open AI Capabilities from Huawei Cloud EI

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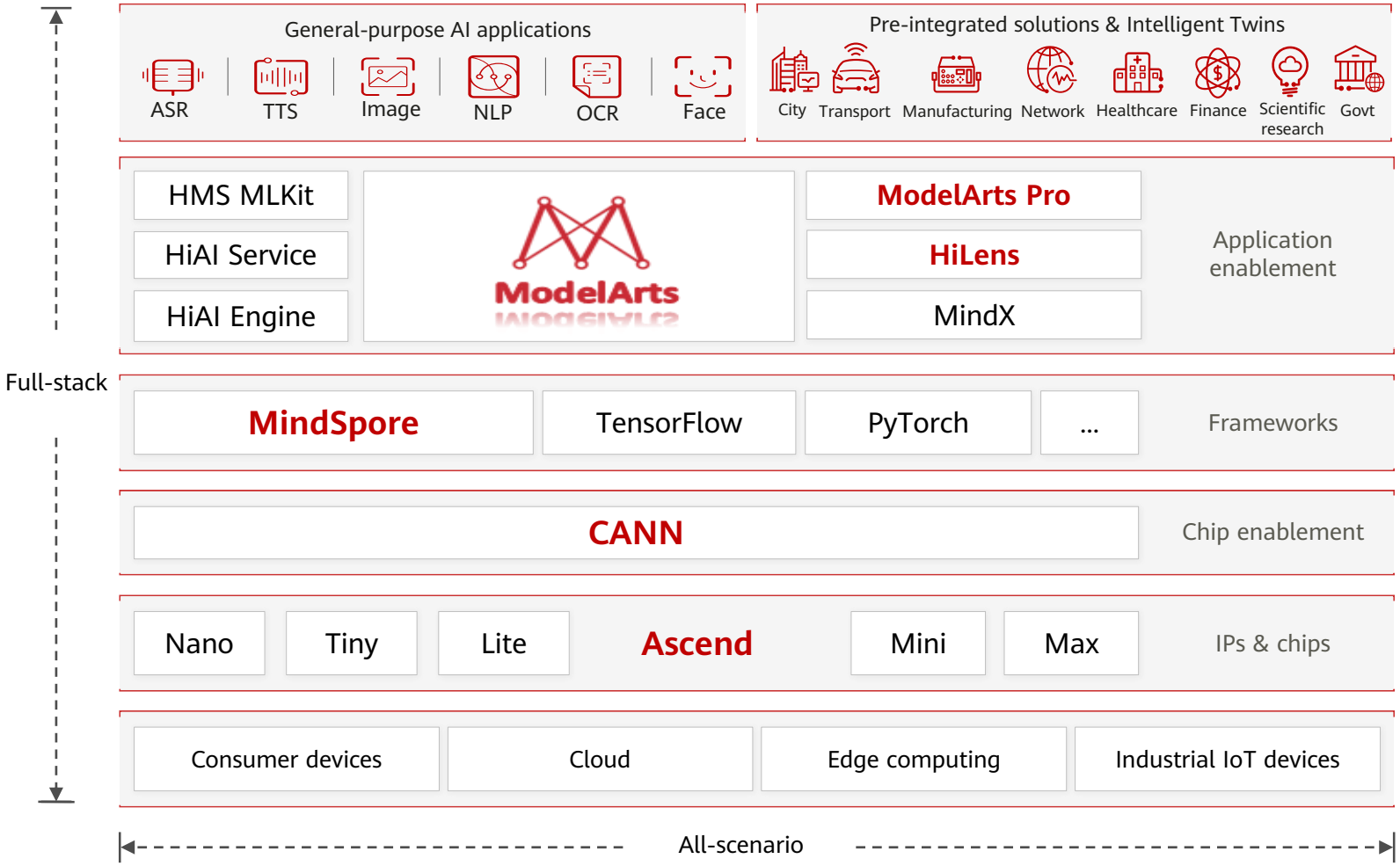
Huawei's Full-Stack, All-Scenario AI Solutions



Application enablement: end-to-end services (**ModelArts**), layered APIs, and pre-integrated solutions.

MindSpore: a unified training and inference framework featuring cloud-edge-device synergy (both standalone and collaborative).

CANN: a chip operator library and highly automated operator development tools.

Ascend: series of AI IPs and chips based on unified and scalable architecture.

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Huawei Cloud EI: Full-Stack, All-Scenario AI Products & Services, Accelerating Intelligent Industry Upgrades

Industries: City | Transport | Electric power | Healthcare | Finance | Network | Scientific research | Meteorology | Water management | Government

Industry ecosystem + Marketplace

DataArts: data enablement

- **Asset center DataArts Hub**
- **Enablement kits DataArts Ekit**
- **Intelligent data lake FusionInsight** (MRS, GaussDB, DataArts, TICS...)

ModelArts: AI enablement

Industry enablement platform AI Cortex | **All-domain sensing engine** | **Decision optimization engine** | **Robot engine** | **Knowledge computing engine**

General AI kits General capabilities

- **General AI capabilities**: NLP/Facial recognition/ Speech recognition
- **Pangu Models**: CV, NLP, multimodal, etc, enabling industrial scale AI development
- **OptVerse AI Solver**: Industry's first commercial AI-powered solver

AI development platform ModelArts (Data preparation, model development, model training, service deployment...)

Full series	Lite	Edge	Standard	Alliance
Multi-architecture compute	CPU	Kunpeng	GPU pool	Ascend pool (NEW)

AI Gallery

- Algorithms
- Models
- Workflows
- ...

MindSpore training and inference framework | TensorFlow | ...

Ascend | GPU | General compute | Kunpeng | x86 and Arm

Public cloud | Hybrid cloud | Intelligent EdgeSite (IES) | Intelligent EdgeFabric (IEF)

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Huawei Cloud ModelArts, a One-Stop AI Development Platform

Huawei Cloud EI Intelligent Twins

General-purpose AI services

Data

AI data framework

Efficient data filtering and automatic data labeling, **100x data preprocessing efficiency**

Algorithm development

Easy-to-setup development environments based on popular frameworks

Simplified model development with the **MoXing library**

Accelerated development with **built-in models/algorithms**

Training

Wizard-based ExeML, **faster**, **no-code model training, no need for AI expertise**

A distributed cluster accelerates training **from weeks to minutes**

Deployment

Device-edge-cloud deployment

Support for **diverse needs**

Ascend-powered inference

Market

A shared AI platform helps customers build internal and external **AI ecosystems**.

AI applications

Workflow data **versioning**, **traceable** development process

ModelArts

AI data framework
100x data processing efficiency

Visualized lifecycle management
Facilitated development

Acceleration with a distributed cluster
Training time reduced from weeks to minutes

Device-edge-cloud deployment
Support for diverse needs

ExeML
Zero-code model development

Shared AI platform
Internal and external AI ecosystem for enterprises

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Huawei Cloud HiLens: A Multimodal AI Application Development Platform Featuring Device-Cloud Synergy

AI application development with device-cloud synergy, fast deployment and iteration, industry-tailored applications

Huawei Cloud ModelArts

Models

HiLens cloud platform

Multimodal/Containerized skills	Video	Image	Speech	IR
Management platform	Development environment: HiLens Studio		Development framework: HiLens Framework	

Devices

Online deployment

HiLens Kit

Atlas500

Atlas800

Ascend 310 + Linux

Offline deployment

3rd-party commercial cameras

Linux / LiteOS

HiSilicon Hi35xx series chips

Device-cloud synergy

Intelligent solutions

- Shopping malls
- Campus
- Homes
- Industrials
- Other

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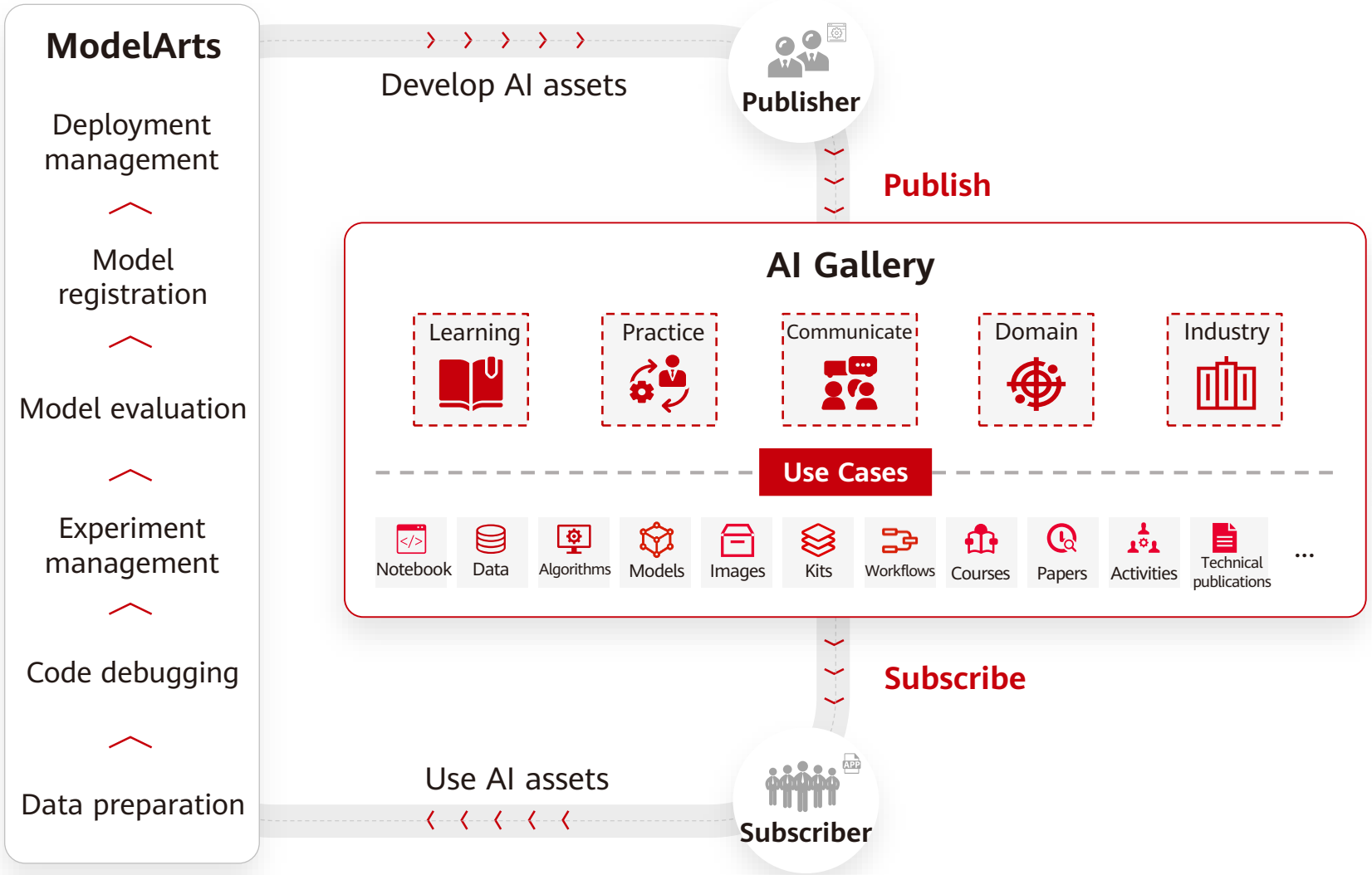
AI Gallery: Bridging Supply and Demand in the AI Ecosystem



...

Pangu Models

Industrial-scale AI development

OptVerse AI Solver

Setting a new bar for operations optimization

RL engines

New choice for intelligent decision making

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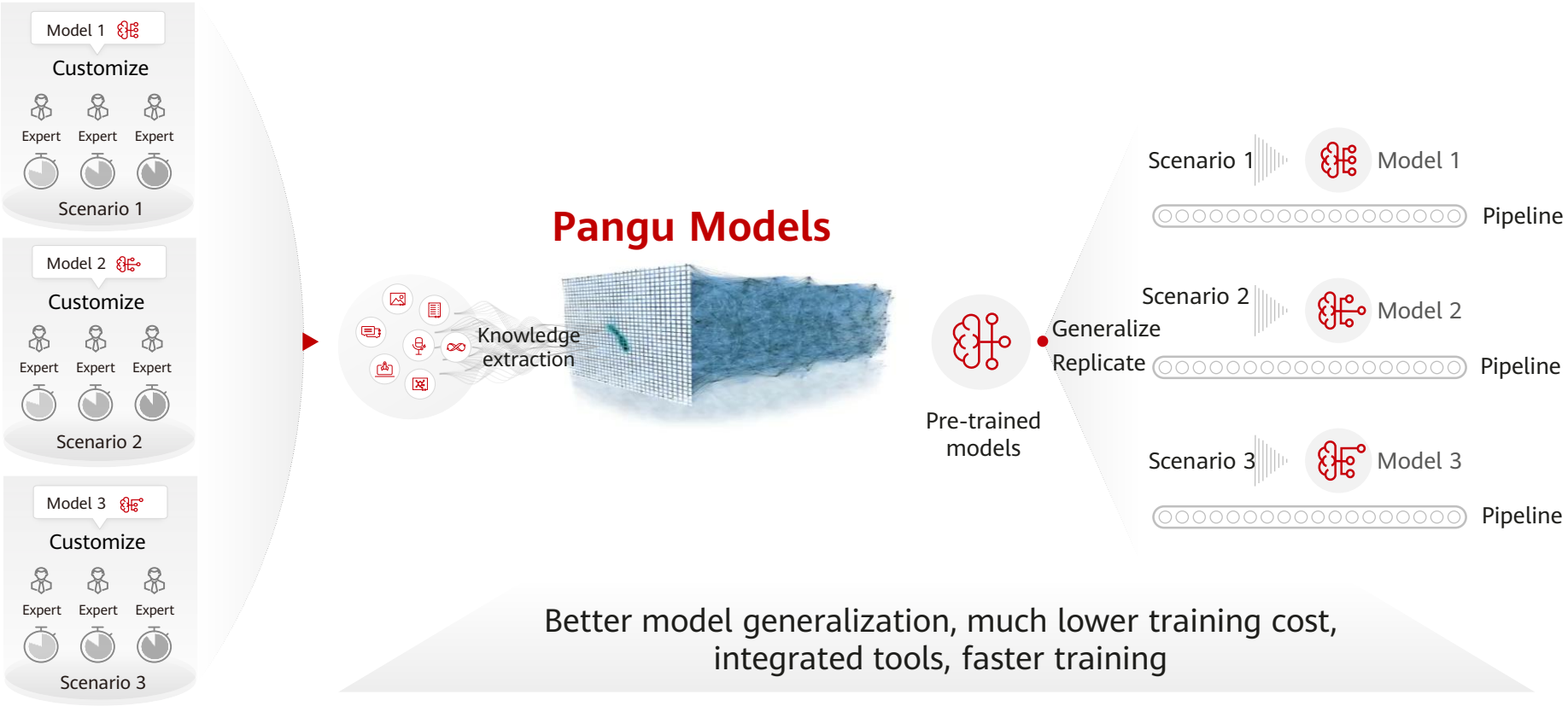
Huawei Cloud Pangu Models: Pioneering A New Paradigm for Industrial Scale AI Development



Siloed AI development ----------▸ **Industrial AI development**

Large network
Hundreds of billions of parameters

Robust network architecture
Overall performance: 10%↑

Excellent generalization
↑10x supported scenarios

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Decision Optimization with Huawei Cloud OptVerse AI Solver

Plan | Schedule | Resource dispatching | Workflow optimization | Improved user experience | Logistics | Inventory optimization | Route optimization

Global optimization

Resource utilization ↑ | Operational efficiency ↑ | Benefits ↑

OptVerse AI Solver

Mathematical programming | Heuristics | **OR**

AI | Machine learning | Deep learning

No. 1 on Hans Mittelmann's Benchmark of Simplex LP Solvers

↑ **30x** Modeling speed
Manual modeling → Intelligent modeling

↑ **30%** Solver efficiency
Manual parameter tuning → AI adaptive tuning

100 million Variables ↑ **10-100x** Computing speed
Single-node computing power → Distributed parallel acceleration

Powering Tianjin Port's intelligent planning platform

[redacted]

Time to develop a plan: **24 hours** → **10 minutes**

Berth allocation plan
5% ↑
Berth utilization

Container crane operations plan
15% ↑
Equipment utilization

Intelligent container yard planning
20% ↓
Container yard reshuffles

Single ship stowage plan
10% ↓
Time in port

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Huawei Cloud CityCore (City Intelligent Twins)

1 Platform + X Engines for Refined City Governance and Public Services

N applications

Public Service for Citizen & Business

- Contactless application
- Unified services
- Intelligent approval
- Intelligent guide
- Intelligent filling
- Remote review

GOV Governance Digitalization

- Smart warning
- Smart supervision
- Digital urban management
- Smart emergency response

Government service hotline

- Smart service tickets
- Smart hotline

Smart heating

- Heat demand prediction
- Substation tuning
- Building heat supply tuning
- Heat network balance

Smart buses

- Precision bus
- Bus scheduling

X engines

Data management engine
Data themes, Data quality, Data diagnosis, Data metrics

All-domain sensing engine
Speech, Text, Video, Image Face

Knowledge computing engine
Knowledge acquisition, Knowledge management, Knowledge modeling, Knowledge application

Robot engine
RPA, Q&A bots, Voice assistant, Quality inspection bots

Scheduling engine
Analysis, Optimization, Scheduling, Simulation

1 platform

- Domain-specific AI development workflows, quick adaptation to a wide range of downstream tasks
- Faster process customization with robotic process automation (RPA)

Smart City AI development & operations platform

- Unified operations and management of AI capabilities, with instant, measurable results
- Open and inclusive for a robust ecosystem

GOV Governance Digitalization

14 days → Handling cycle → 3 h

200+ events — Automatic dispatch

40+ events — Intelligent discovery

Smart government hotline

Call completion rate: 20% ↑

200k+ service tickets — AI-supported correlation analysis

1,000+ topics — AI extraction

Public Service for Citizen & Business

60% ↓ Less paper

40% ↓ Faster handling time

Smart heating

97% On-demand production accuracy

12.1% ↓ Average energy consumption

Smart traffic

Intersection overflows 20% ↓

18% ↑ Average vehicle speed

10-15m ↓ Morning peak hours

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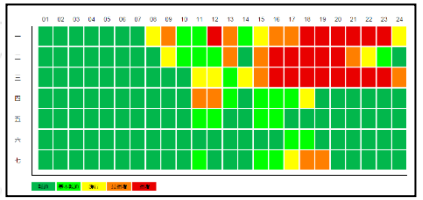
Huawei Cloud Smart Traffic Solution — TrafficGo

Traffic knowledge graphs, recognizable patterns

- Road capacity precision profiling
- Traffic pattern precision profiling

Road health archives based on multi-source, converged data, and quantitative analysis of congestion causes

- Traffic condition analysis
- Traffic situation analysis



2. Understand

3. Diagnose

1. Sense

4. Optimize

5. Evaluate

All things sensing, complete, digitized traffic information

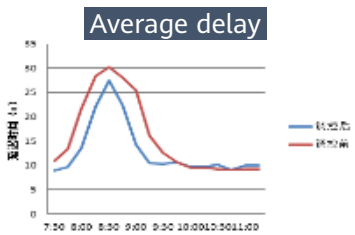
- Signal controller
- ITS camera
- Intelligent intersection
- Internet data

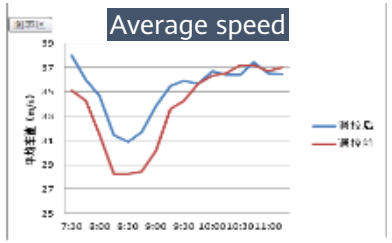
Dynamic optimization by scenario, time, and location

- Intelligent signal optimization
- Traffic organization optimization
- Visual command

Multi-dimensional traffic metric systems, evaluation of iterative optimizations

- Traffic safety evaluation
- Traffic efficiency evaluation
- Traffic order evaluation
- Traffic environment evaluation
- Economic evaluation





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Smart Traffic — Traffic Condition Analysis

~10%
> 30s

~100%
1 second

Road-level vehicle flow

Lane-level
Vehicle-level

Single metric Road network-level

Multi-metrics
Road network-level

Deduction based on samples: incomplete, inaccurate

Traffic condition analysis based on Internet or sensor data, road conditions deduced using traffic models

Digital re-creation based on comprehensive, real-time data: more accurate, more objective

Intelligent analysis of data collected from intelligent intersections and existing ePolice cameras, road conditions accurately calculated based on complete vehicle passage records

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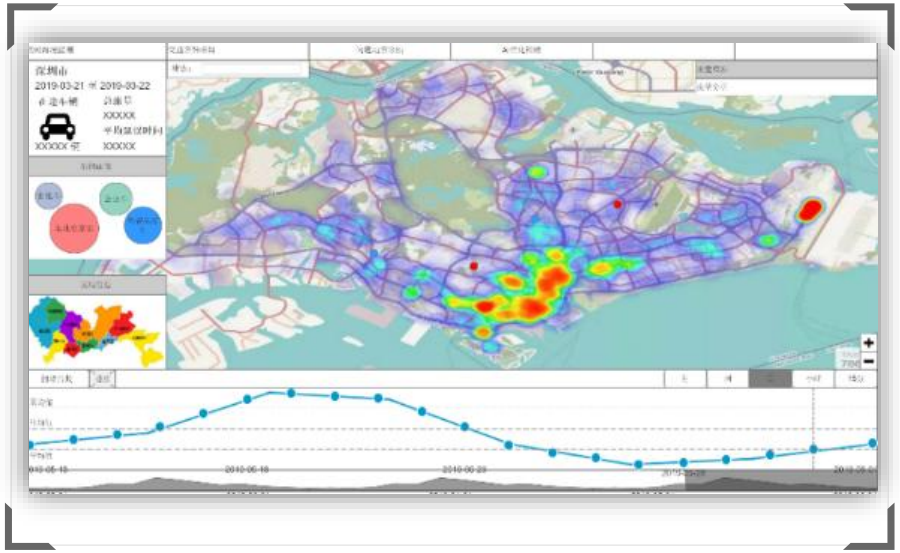
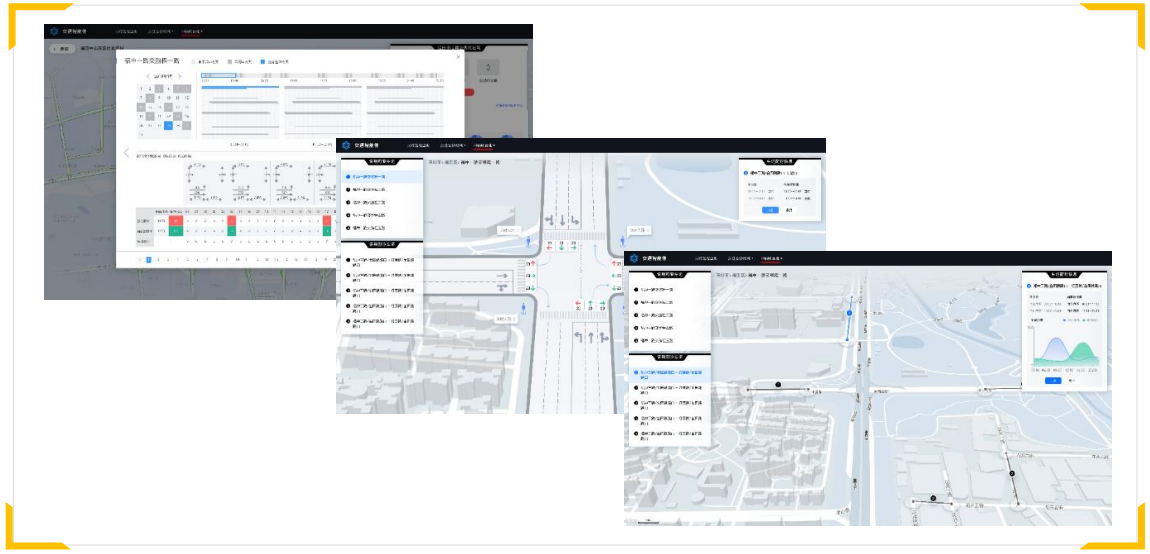
Smart Traffic — Traffic Situation Evaluation





Superficial trend analysis

Predict trends based on superficial info

Combine historical data with current conditions to predict future trends

Dynamic situation evaluation

In-depth analysis discovers the root causes of congestion

Combine real-time road conditions with people's travel needs

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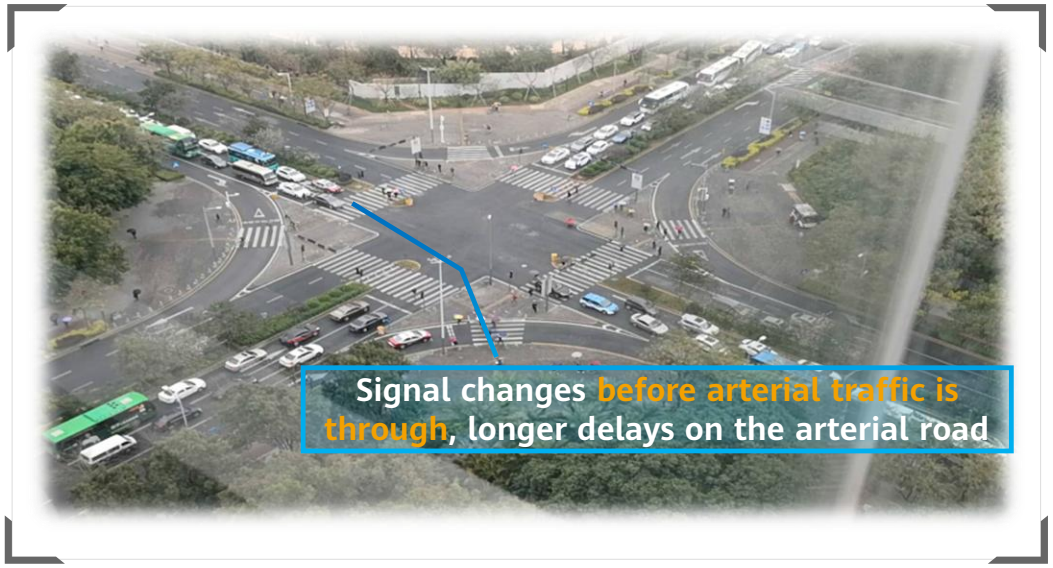
Smart Traffic — Intelligent Signal Optimization

Before: drivers count the time before light changes



- Collecting traffic data using magnetic sensors: incomplete, inaccurate data, takes time and space to deploy the sensors
- Signal timing optimization relies on human experience, incapable of dynamic adaptation or coordinated optimization.

Now: Traffic lights count vehicles, then change

- **Complete data collection at intelligent intersections**: multi-dimensional data and events like queuing, vehicle flow, overflow, and congestion.
- **Adaptive control: reinforcement learning**, coordinated optimization, optimized policies for complex scenarios, real-time update of decision-making models; signal timing solution based on motor vehicles, non-motor vehicles, as well as pedestrians
- **20% less queue overflow** at intersections

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Smart Traffic — Transportation Organization Optimization

Accurate vehicle route data at intelligent intersections helps improve the design of intersections for higher efficiency.

- Intersection broadening
- Channelization
- Lane rearrangement
- More recognizable tidal lanes

Heavy dependence on human expertise

Traffic organization optimization mainly performed offline, inefficient, a waste of expert resources.

Human surveys take a long time and cannot guarantee data accuracy.

Real-time, accurate traffic data + expert models

Able to accurately calculate and aggregate travel needs to support quick optimization.

Precision profiling of traffic flows, optimization carefully designed based on simulations

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Huawei Cloud Industrial AI Cortex (Intelligent Twins)

Coking coal blending

- Expert experience
- Auto-learning of multiple parameters
- E2E benefit calculation

Quality prediction accuracy > 95%

Savings per million tons of coke CNY20+ million

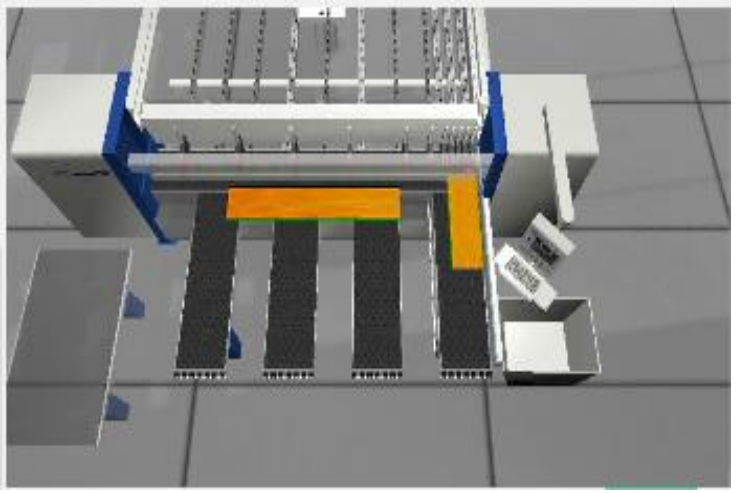
Steelmaking ingredients

Composition prediction accuracy

85%-90% → 95%

Alloying costs saved

CNY20 million/year



Cutting

Higher board utilization

84%-86% → 86%-88%

Plate cost reduction

CNY40 million/year

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Huawei Cloud Healthcare Solution Overview

Ministry of Health | Healthcare organizations | Research institutes | DNA sequencing companies | Pharmaceutical companies

DNA sequencing

Drug R&D

Imaging cloud

AI-assisted diagnosis

Big data	**Database**	**AI**	**IoT**	**EIHealth**
Compute	**Storage**	**Networking**	**Security**	**O&M**

200+ Huawei Cloud service capabilities

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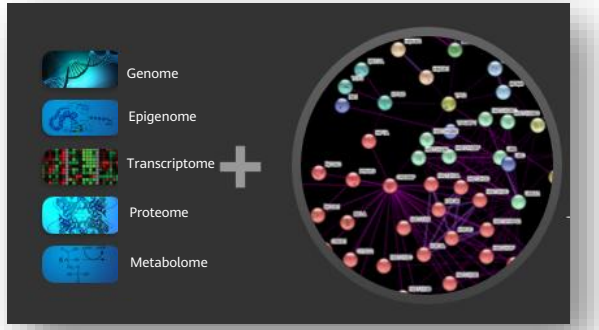
Huawei Cloud EIHealth Solution

Full-stack, open, and professional healthcare AI R&D platform

· Gene Sequencing ·

Single- and multi-omics, gene network, and gene sequencing



Genome engine

AutoGenome
World's first auto genome modeling tool

AutoOmics
Brand-new multi-omics AutoML framework

· Drug R&D ·

Drug generation optimization, protein structure prediction, molecular dynamics simulation, federated learning, and more

Drug R&D engine

100M parameters
Pangu Drug Molecule Model

20+ tasks
SOTA top performance

Huawei Cloud EIHealth

Customer Highlights ▸ **Limited technologies** → **Full digital sovereignty Total control**

Years → **1 month**
Lead discovery

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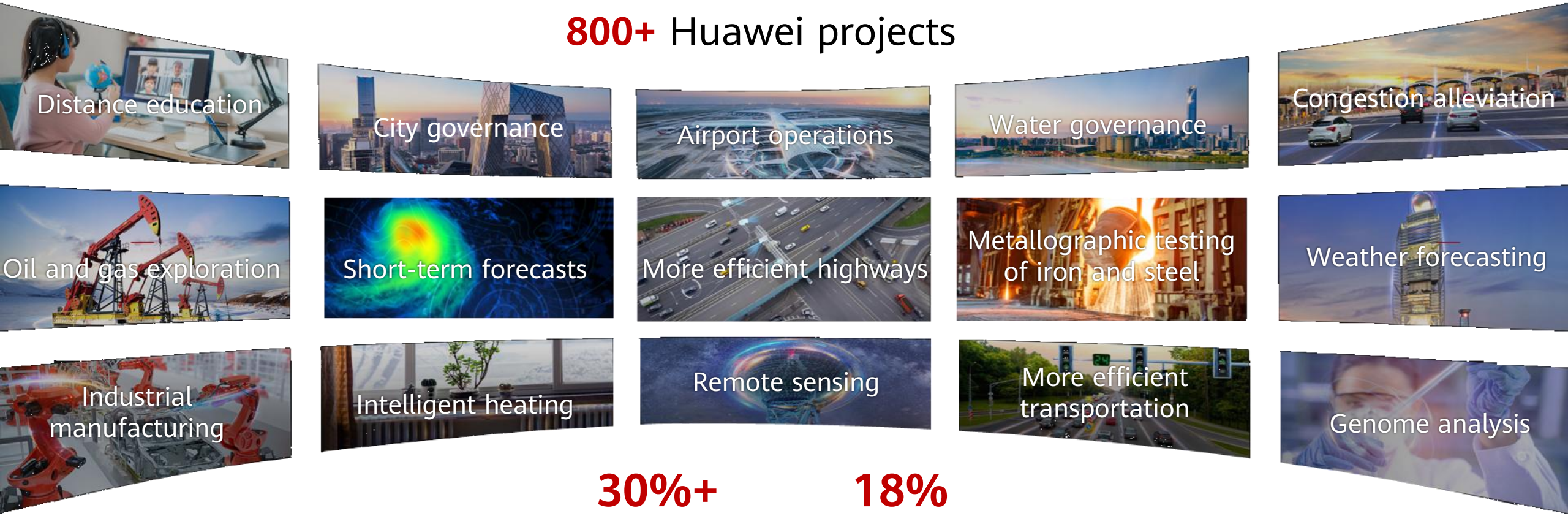
Huawei Cloud AI Explores Industry Best Practices with Customers and Partners

800+ Huawei projects



30%+

core business systems are powered by AI

18%

higher profitability

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Section Summary

- In Huawei's full-stack, all-scenario AI solutions, "full-stack" covers IPs and chips, chip enablement, frameworks, and application enablement, and "all-scenario" covers consumer devices, cloud, edge computing, and industrial IoT devices.
- Huawei Cloud supports developers with open AI capabilities, including general APIs, ModelArts, HiLens, AI Gallery, and industry-specific AI solutions, helping to build a robust AI ecosystem that contributes to an intelligent world.

Q&A

1. (Multiple-answer question) Which of the following are included in the term "full-stack" in Huawei's full-stack, all-scenario AI solutions? ()

A. IPs and chips

B. Chip enablement

C. Frameworks

D. Application enablement

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Thank You.

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